

KRISHEN MATHUR

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Education

University of Illinois at Urbana-Champaign

Expected December 2027

B.S. Electrical Engineering, Minor in Semiconductor Engineering GPA: 3.50

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- **Relevant Coursework:** Power Electronics, Analog Signal Processing, Digital/Analog Circuits, Semiconductor Electronics, Computer-Aided Design (PCB), Computer Systems and Programming, Probability and Statistics

Experience

Power Electronics Intern – Sourcing & Development

Summer 2026

Centre for Innovation, Technology and Learning (CITL) – Hero MotoCorp

Jaipur, India

- Simulated an active **PFC boost converter** in **LTspice** for a 580W onboard charger, reaching **94% simulated efficiency** with two-loop average current mode control.
- Characterized converter behavior under startup, inrush, and load-step conditions to size protection components and confirm control loop stability across the operating range.
- Mapped the charger's full power path (EMI filtering, rectification, PFC, LLC resonant stage, isolation, control) and audited SMT assembly lines to trace design intent through to production.
- Built a **120+ line-item EBOM** spanning power MOSFETs, magnetics, connectors, potting, and enclosure, supporting sourcing decisions and vendor negotiations.
- Authored a concept proposal for a **580W wireless charging system**, evaluating coil coupling, misalignment tolerance, and thermal limits for two-wheeler EVs.

Team Lead, Student Technology Consultants

Mar 2025 – Present

University of Illinois at Urbana-Champaign – Technology Services

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- Lead **30+ consultants** handling **500–800 tickets weekly**; own scheduling, escalation paths, and new-hire training on diagnostic workflows.
- Resolve escalated hardware, networking, and AV failures across classroom and event spaces, driving from reported symptom to root cause under live time pressure.

Projects

Analog DJ Mixing Board

May 2026 – Present

Personal Project | Altium, LTspice, Python

- Designing a 2-channel analog mixer (op-amp gain stages, summing amplifier, 3-band EQ, crossfader) from **schematic capture through fabricated multi-layer PCB** in **Altium**.
- Laid out dedicated ground planes, star grounding, and local decoupling to suppress channel crosstalk and supply-coupled noise.
- Validating each module on the bench with oscilloscope, function generator, and DMM, comparing measured gain, frequency response, THD, and noise floor against **SPICE** targets.
- Automated the measurement-to-Bode-plot workflow in **Python** to track response drift across board revisions.

Project Stealth – Wearable EMG & IMU System

Aug 2025 – Present

I-MADE | Arduino Nano ESP32, C++, Python, BLE

Champaign, IL

- Engineered a 2-channel EMG + IMU acquisition system (MyoWare 2.0, MPU6050) streaming synchronized sensor data over **BLE** to a live Python pipeline.
- Designed the analog front end with instrumentation amplification and 20–450 Hz active band-pass filtering to isolate the surface-EMG band.
- Performed **board bring-up and debug**, isolating sensor saturation, ground-loop noise, and intermittent BLE dropouts to recover clean multi-channel capture.
- Trained an SVM classifier on filtered EMG features for real-time gesture detection, driving a live game demo for **300+ attendees** at Engineering Open House and a pitch at the Cozad New Venture Challenge.

Technical Skills

Power Electronics: PFC & Boost Converters, DC-DC Topologies, LLC Resonant Conversion, Average Current Mode Control, EMI Filtering, Magnetics Selection

Hardware & PCB: Analog Circuit Design, Schematic Capture, Multi-Layer PCB Layout (Altium), Op-Amp Circuits, Signal Conditioning, Grounding & Decoupling, Soldering

Test & Bring-Up: Board Bring-Up, Hardware Debug, Oscilloscope, Function Generator, DMM, Frequency Response & THD Characterization, SPICE Simulation

Embedded & Software: C, C++, Python (signal processing, data analysis), Assembly, Embedded Firmware, Serial & BLE Communication, Git/GitHub, Linux, LaTeX, Fusion 360

Interests: Strength Training, Electronic Music Production, Soccer, Billiards, Table Tennis